

Power Analysis for Children's Language Ability Study

Xiaohan Shi, Yunxiang Zhang, Gary Wang

Brief Introduction

- Client: Malvika Khandelwal
- Data collection: design an experiment to evaluate children's language learning skill
- Objective: Determine the minimum sample size of children with disorder. She wants to achieve an effect size of 0.8 between healthy vs disordered kids by using the power analysis.
- Current data: 41 healthy children, ideally 25 disorder children

Client Model

Correct $\sim$ Age + Language_ability_score + Disorder

We start with:

Correct $\sim$ Disorder

- Effect size = Odds Ratio = 0.8, a binary intervention variable with a binary outcome.
 - The odds of answering correctly in the disordered group are only 80% as high as the odds in the healthy group.
- Power = 0.8 (goal)

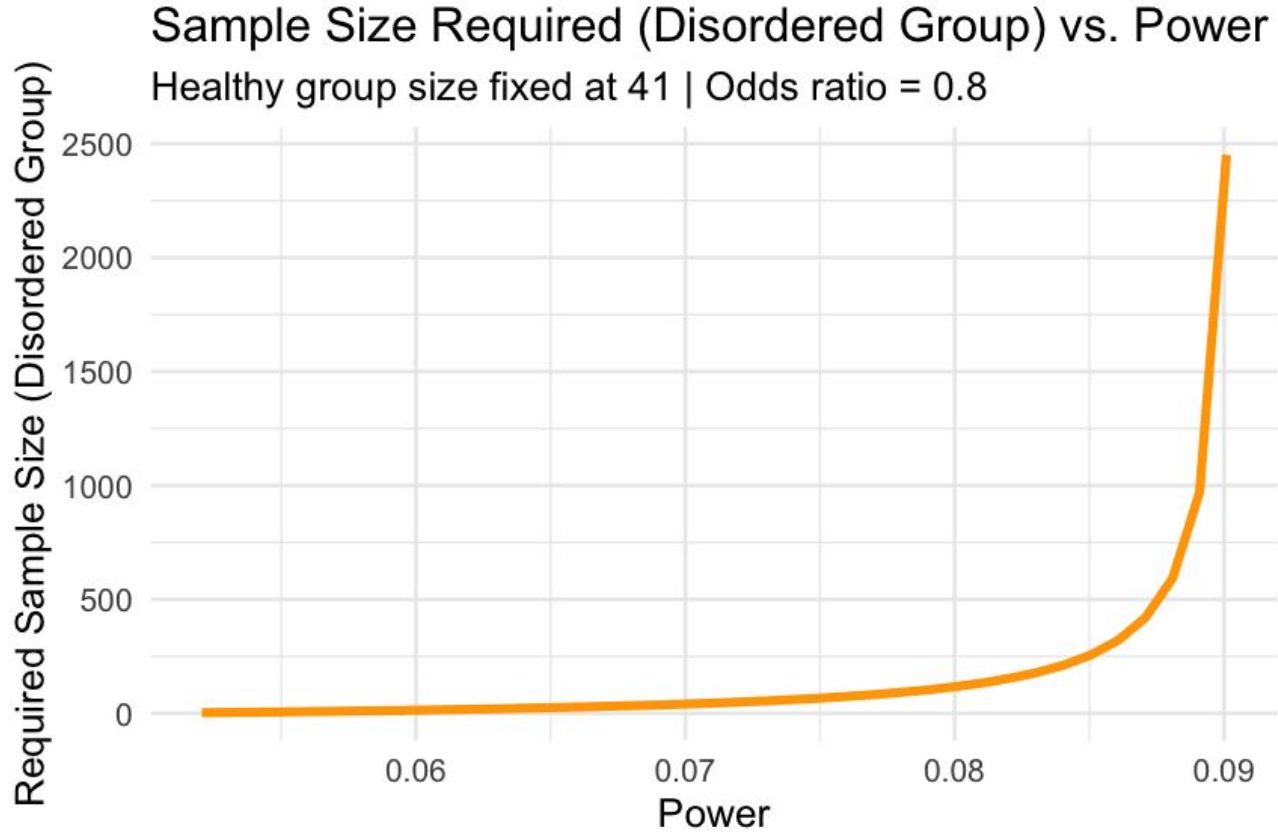
Code

```
# set parameter
p1 <- 0.7          # No disorder group success probability 70%
OR <- 0.8          # Effect size (OR=0.8, coef.=log(0.8))
new_p2<-plogis(logit(p1)-log(OR))
new_p2
h <- ES.h(p1, p2) | # Cohen's h for two proportions

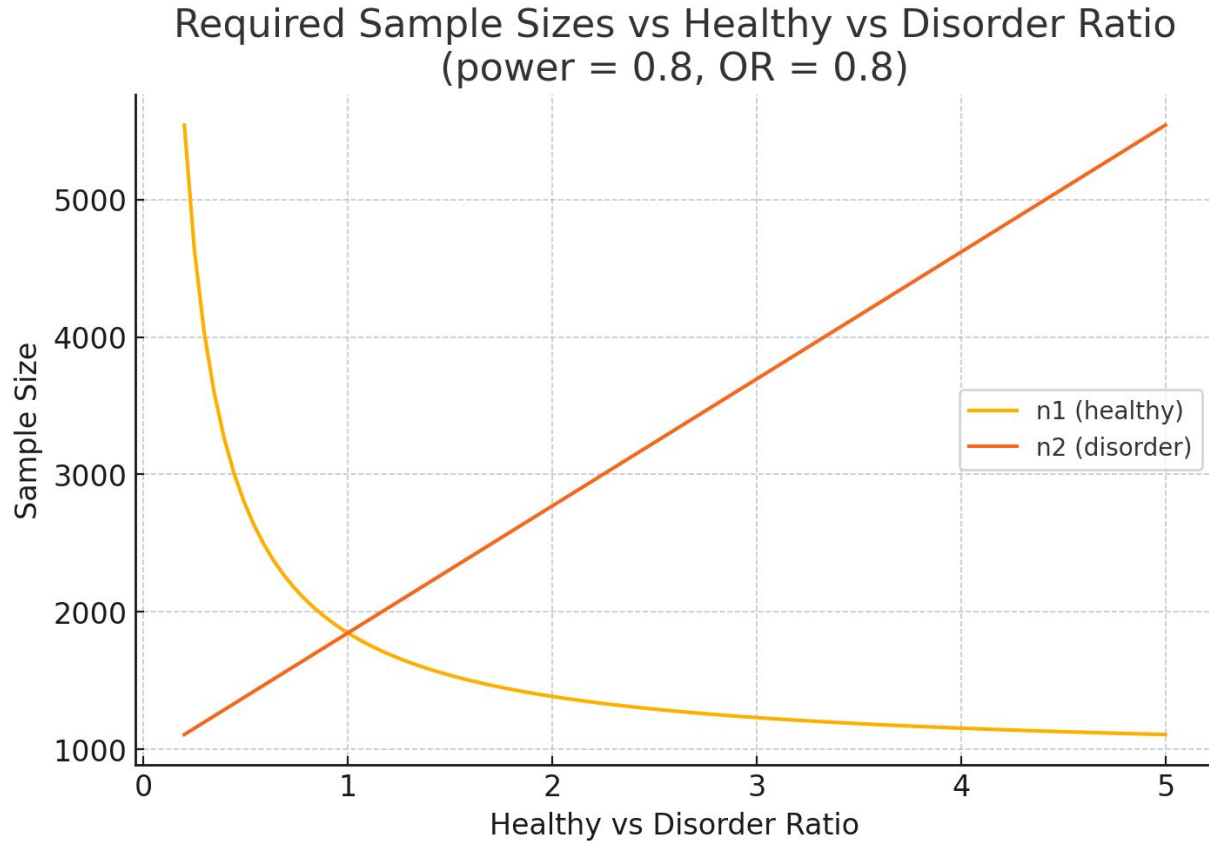
n_control <- 41

# calculate disorder prople
calculate_n_disorder <- function(target_power, max_disorder = 10000) {
  for (n_disorder in 10:max_disorder) {
    result <- pwr.2p2n.test(h = h, n1 = n_disorder, n2 = n_control, sig.level = 0.05)
    if (result$power >= target_power) {
      return(n_disorder)
    }
  }
  return(NA)
}
```

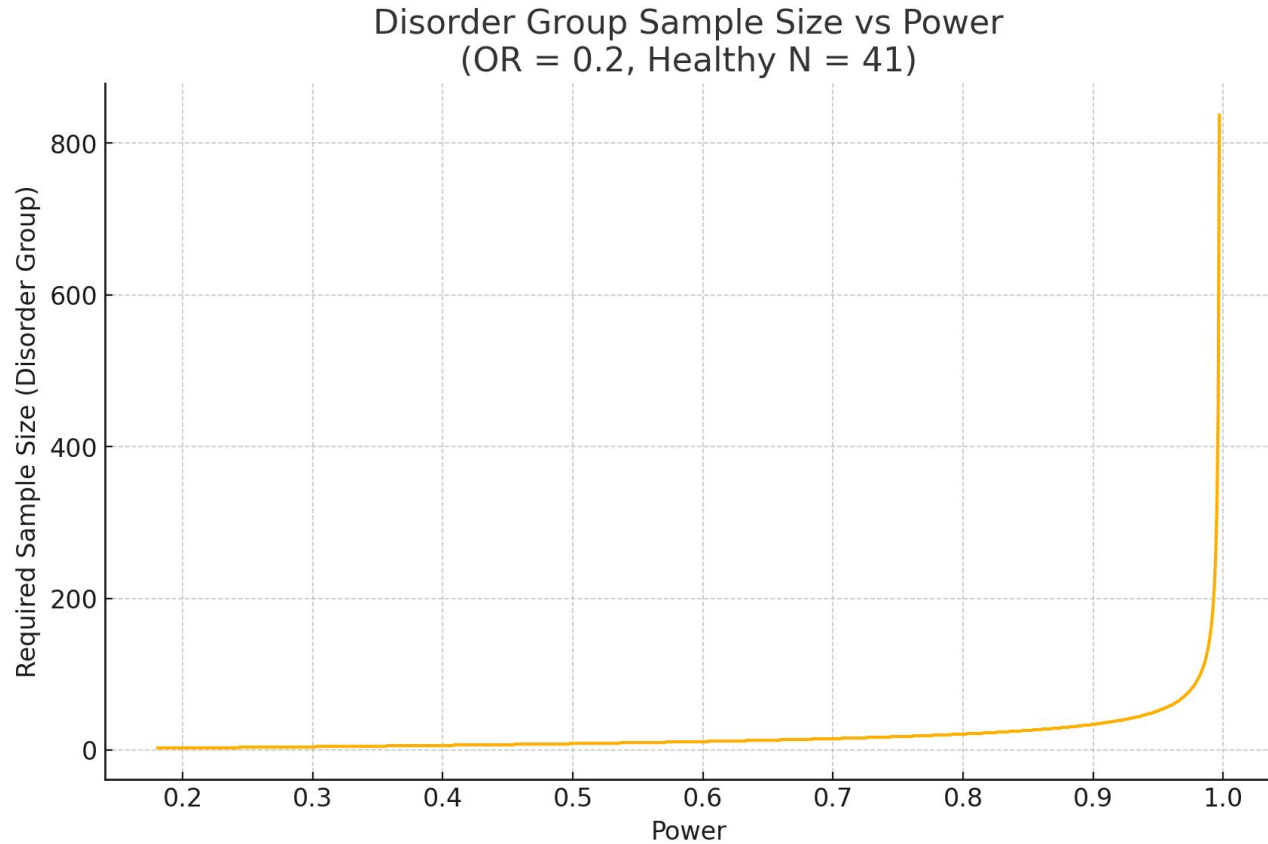
Result



What if



What if



Discussion

Suggestion: Lower the expected odds ratio to achieve a large effect between the two groups

Sample bias: Age

- Age may be imbalanced in Healthy vs Disorder
- Older children are more likely to answer correctly and be considered as have no disorder